BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: 8473

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics - Paper: II - Objective (ii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8473

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. Energy of electron in the infinite orbit of hydrogen atom is:

(A) 13.2 eV
(B) -13.2 eV
(C) 3.4 eV
(D) Zero

Q2. Photoelectric effect is explained by considering light as:

(A) Electromagnetic waves
(B) Corpuscles
(C) Wave front having energy
(D) Simple waves

Q3. Energy of a photon is independent of:

(A) Intensity of light
(B) Frequency of light
(C) Wavelength of light
(D) Velocity of light in a medium

Q4. Current gain of transistor is of the order of:

(A) Decimal
(B) Hundreds
(C) Thousands
(D) 10

Q5. A photo diode can turn its current ON and OFF in:

(A) Nano second
(B) Milli second
(C) Seconds
(D) 100 second

Q6. The number of different crystal systems based on the geometrical arrangement of their atoms is:

(A) 5
(B) 7
(C) 4
(D) 9

Q7. Power factor in resistive circuit when AC is passing will be:

(A) Zero
(B) 1
(C) $1/\sqrt{2}$
(D) 1/2

Q8. Power dissipation in pure inductor circuit over the cycle is:

(A) VI
(B) VI $\cos\theta$
(C) V^2/R
(D) Zero

Q9. Efficiency of practical transformer is less than ideal one due to:

(A) Eddy current
(B) High current
(C) Low current
(D) Low voltage

Q10. VSA^{-1} is called:

(A) Joule
(B) Watt
(C) Henry
(D) Newton

Q11. A one coulomb charge of mass one gram is in electric field of $1NC^{-1}$, acceleration will be:

(A) 100 ms^{-2}
(B) 1000 ms^{-2}
(C) 1 ms^{-2}
(D) 10 ms^{-2}

Q12. Magnetic flux through area $5m^2k$ due to magnetic field 3i + 2j tesla is:

(A) 15 Wb
(B) 10 Wb
(C) 30 Wb
(D) Zero

Q13. A wire of length l and resistance R is cut into three equal pieces and twisted. Equivalent resistance will be:

(A) R
(B) R/3
(C) R/9
(D) R/4

Q14. Shape of Gaussian surface should be:

(A) Closed
(B) Spherical
(C) Circular
(D) Box type

Q15. Toner is given:

(A) Positive charge
(B) Negative charge
(C) Conventional current
(D) No charge

Q16. It is very difficult to dispose off radioactive waste due to:

(A) Long half life
(B) High energy
(C) Uncontrolled chain reaction
(D) Fast chemical reaction

Q17. Size of quark is of the order of:

(A) Less than 10^{-15} m
(B) Less than 10^{-10} m
(C) Less than 10^{-9} m
(D) Less than 10^{-18} m

Physics - Paper: II - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. What are the factors upon which electric flux depend?

ii. Define Electron Volt. Prove that $1eV = 1.6\times10^{-19}J$.

iii. Suppose that you follow an electric field line due to a positive point charge. Do electric field and the potential increase or decrease?

iv. Do electron tend to go to the region of high potential or low potential?

v. Can an electron at rest be set in motion with a magnet?

vi. How can you use a magnetic field to separate isotopes of chemical element?

vii. If a charge particle moves in a straight line through some region of space, can you say that magnetic field in that region is zero?

viii. Write down main parts of CRO.

ix. What are Hardons and Leptons? Explain with examples.

x. What is meant by dose of radiation? Give its unit.

xi. If someone accidentally swallows an α-source and β-source which would be more dangerous to him? Explain why?

xii. A particle which produces more ionization is less penetrating. Why?

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Distinguish between the conventional current and electronic current.

ii. How can you use a rheostat as a variable resistor in a circuit?

iii. Why does the resistance of a conductor rise with temperature?

iv. An AC voltmeter reads 250V. What is its peak value?

v. Define Modulation with its types.

vi. How many times per second will an incandescent lamp reach maximum brilliance when connected to a 50Hz source?

vii. Define Curie Temperature. What is curie temperature for iron?

viii. Differentiate between donor atoms and acceptor atoms.

ix. Define the terms (a) Elastic limit (b) Yield point

x. Define Logic Gates.

xi. Why ordinary silicon diodes do not emit light?

xii. The base current in a transistor is very small. Why?

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. Does the induced emf in a circuit depend on the resistance of the circuit? Does the induced current depend on the resistance of the circuit?

ii. Does the induced emf always act to decrease the magnetic flux through a circuit?

iii. Write two methods for determining the induced emf in a loop.

iv. State Faraday's law of electromagnetic induction and write its mathematical expression.

v. What happens to total radiation from a black body if its absolute temperature is doubled?

vi. Why do we not observe a Compton effect with visible light?

vii. Distinguish between general and special theory of relativity?

viii. Distinguish between stimulated and spontaneous emission.

ix. What is meant by line spectrum? Explain how line spectrum can be used for identification of elements?

> Note: SECTION-C (Each question carries Eight (8) Marks)

Q5.. (Marks 5) (a) What is Wheatstone bridge? How it can be used to find the unknown resistance? (b) A particle having a charge of 20 electrons on it falls through the potential difference of 100 volts. Calculate the energy acquired by it in electron volt(eV).

Q6.. (Marks 8) (a) Describe how charge to mass (e/m) ratio of an electron can be determined by projecting it perpendicular to a magnetic field. 5 (b) Two coils are placed side by side. An emf of 0.8V is observed in one coil when the current is changing at the rate of $200As^{-1}$ in the other coil. What is the mutual inductance of the coils? 3

Q7.. (Marks 8) (a) Describe the A.C through a R-C series circuit. 5 (b) Calculate gain of non-inverting amplifier as shown in given figure. 3

Q8.. (Marks 8) (a) Define Compton Effect and find the expression for Compton shift. Draw its scattering diagram and label it. 5 (b) 1.25cm diameter cylinder is subjected to a load of 2500 Kg. Calculate the stress on the bar in mega pascals. 3

Q9.. (Marks 8) (a) What is nuclear fission? Explain fission chain reaction in detail. 6 (b) The wavelength of K X-ray from copper is 1.377×10^{-10} m. What is the energy difference between the two levels from which this transition results? 3

GROUP – II

Physics - Paper: II - Objective (ii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8474

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Note: SECTION-A

Q1. At high frequency the value of reactance of a capacitor in A.C circuit will be:

(A) Small
(B) Zero
(C) Large
(D) Infinite

Q2. The mean value of A.C is:

(A) 1
(B) 0
(C) I_o
(D) $I_o/\sqrt{2}$

Q3. Torque is produced in a current carrying coil when it is placed in a:

(A) Electric field
(B) Magnetic field
(C) Gravitational field
(D) Nuclear field

Q4. The application of mutual induction is a:

(A) D.C motor
(B) Radio
(C) Television
(D) Transformer

Q5. Energy density in inductor is given by:

(A) $(1/2)(B^2/\mu_0)$
(B) $(1/2)(B^2/\mu_0^2)$
(C) $B^2/(2\mu_0^2)$
(D) $B^2/(2\mu_0)$

Q6. Which one has least resistance?

(A) Galvanometer
(B) Ammeter
(C) Voltmeter
(D) Ohmmeter

Q7. When a charge of 5μC passes through a conductor in 2 sec. the current in conductor is:

(A) 10A
(B) 2.5A
(C) 2.5mA
(D) 2.5μA

Q8. The electric intensity due to an infinite sheet of positive charge is:

(A) $E = \sigma/2\varepsilon_0$
(B) $E = \sigma/\varepsilon_0$
(C) $E = 2\sigma/\varepsilon_0$
(D) $E = 1/2\sigma\varepsilon_0$

Q9. Which one is photoconductor?

(A) Copper
(B) Selenium
(C) Mercury
(D) Aluminium

Q10. The binding energy per nucleon is maximum for:

(A) Helium
(B) Iron
(C) Polonium
(D) Radium

Q11. Half life of Uranium -239 is:

(A) 26.5 minutes
(B) 24.5 minutes
(C) 25.5 minutes
(D) 23.5 minutes

Q12. Photons emitted in inner shell transition are:

(A) Continuous X-rays
(B) Discontinuous X-rays
(C) Characteristics X-rays
(D) Energetic X-rays

Q13. The value of Wien's constant is:

(A) 2.9×10^{-3} mk
(B) 2.9×10^{3} mk
(C) 2.9×10^{-5} mk
(D) 2.9×10^{4} mk

Q14. Platinum wire becomes yellow at a temperature of:

(A) 900°C (B) 500°C

(C) 1100°C (D) 1300°C

Q15. A device which used for the conversion of A.C into D.C is called:

(A) An oscillator (B) Detector

(C) An amplifier (D) Rectifier

Q16. The value of potential barrier for Silicon at room temperature is:

(A) 0.3 V (B) 0.7 V

(C) 0.5 V (D) 0.9 V

Q17. The S.I unit of stress is same as that of:

(A) Force (B) Pressure

(C) Momentum (D) Work

Physics - Paper: II - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. What is the function of drum in photocopier?

ii. Prove that electric intensity and potential gradient has same unit.

iii. Do electrons tend to go to region of high potential or of low potential?

iv. Electric lines of force never cross. Why?

v. How direction of magnetic field is determined in a current carrying long wire?

vi. Prove that $NA^{-1}m^{-1} = Wbm^{-2}$. Justify your answer.

vii. Why the resistance of an ammeter should be very low?

viii. Is it possible to orient a current loop in a uniform magnetic field such that the loop will not tend to rotate?

ix. Which radiation dose would deposit more in the body (a) 10mGy to hand or (b) 1mGy to entire body?

x. What do you understand by background radiation? Write two sources of these radiation.

xi. What will be the charge on bottom, bottom and bottom quark combination baryon?

xii. Differentiate between thermal and fast reactors.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. A potential difference is applied across the ends of a copper wire. What will be effect on the drift velocity of free electrons by (a) increasing potential (b) decreasing the length and the temperature of the wire?

ii. Why does the resistance of a conductor rise with temperature?

iii. Differentiate between the terminal potential and emf on the basis of open and closed circuit.

iv. How does doubling the frequency affect the reactance of (a) an inductor (b) a capacitor?

v. What is meant by A.M and F.M?

vi. A sinusoidal current has peak value of 14.14A. what will be its rms value?

vii. Explain, why the ball comes to its original size when stress is removed?

viii. Differentiate between steel and soft iron on the basis of hysteresis loop area.

ix. Distinguish between crystalline, amorphous and polymeric solids.

x. What is the net charge on n-type or p-type substance?

xi. Why the base current in a transistor is very small?

xii. What are sensors? Explain how Light Dependent Resistance (LDR) works as sensor of light.

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. On what factors the self-inductance of a coil depends?

ii. What is the principle of an electric generator?

iii. Four unmarked wires emerge from a transformer. What steps should you take to determine the turns ratio?

iv. How would you position a flat loop of wire in a changing magnetic field so that there is no emf induced in the loop?

v. Why do not we observe Compton Effect with visible light?

vi. Can pair production take place in vacuum? Explain.

vii. State uncertainty principle.

viii. How X-rays cause damage to living tissue?

ix. How can the spectrum of Hydrogen contain so many lines when Hydrogen contains only one electron?

> Note: SECTION-C (Each question carries Eight (8) Marks)

Q5.. (Marks 5) (a) Find the charge on an electron by Millikan's method. (b) A rectangular bar of iron is 2cm by 2cm in cross-section and 40cm long. Calculate its resistance if the resistivity of iron is 11×10^{-8} Ωm

Q6.. (Marks 8) (a) Derive the expression for force on moving electric charge in a uniform magnetic field. Also determine its direction. 5 (b) A square coil of side 16cm has 200 turns and rotates in uniform magnetic field of magnitude 0.05T. If the peak emf is 12V. What is the angular velocity of the coil? 3

Q7.. (Marks 8) (a) How power is calculated in A.C circuit? Draw circuit diagram for RLC series resonating circuit. Discuss the behaviour of this circuit for A.C and also write down its properties. 5 (b) The current flowing into the base of a transistor is 100 μA. Find its collector current Ic, its emitter current IE and the ratio Ic/IE, if the value of current gain β is 100. 3

Q8.. (Marks 8) (a) What is meant by strain energy? How can it be determined from force extension graph? 5 (b) Assuming you radiate as does a black body at your body temperature 37°C. At what wavelength do you emit the most energy? 3

Q9.. (Marks 8) (a) Discuss the nuclear fission reaction in detail. 5 (b) Compute the shortest wavelength radiation in the Balmer series. What value of n must be used? 3